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Does Playing Surface Affect the Risk of Anterior Cruciate Ligament Injuries in the National Football League? A Look at the 2017 - 2021 NFL seasons

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Background: Anterior cruciate ligament (ACL) injuries are devastating injuries for athletes. Prior studies have shown increased ACL injury rates on non-natural surfaces versus natural grass in several sports. The purpose of this study is to calculate the prevalence of ACL injuries in the NFL on natural versus non-natural surfaces to determine if there is a significant increase on non-natural surfaces.

Methods: Accessing publicly available data for NFL seasons beginning with the 2017-2018 season through 2021-2022 seasons, all ACL injuries with publicly available data concerning timing and playing surface were recorded and categorized according to playing surface. Practice injuries or those without an identifiable playing surface were excluded. Incidence rates, defined as ACL ruptures per game, were calculated. ACL injuries were recorded for each playing surface, as well as the combined category of non-natural grass surface. Odds ratio was calculated to compare the risk of ACL rupture on non-natural surfaces vs natural grass.

Results: During the 2017-2021 NFL seasons, 173 ACL ruptures were identified with known surfaces. Injury rate for non-natural surfaces was 0.134 compared to 0.097 for grass. Injury rate ratio for non-natural vs natural grass surfaces was 1.211, a 21.1% increased risk of ACL injury in the NFL on non-natural surfaces vs natural grass. OR for non-natural surfaces 1.239 (95% CI 0.900-1.704). Based on these findings there is a trend toward increased risk of ACL injury on non natural grass surfaces, however this did not reach statistical significance.

Conclusion: Numerous published studies show trends towards increasing rates of ACL injuries on non-natural playing surfaces vs natural grass. Based on our findings the difference is not statistically significant, however it does trend toward increased risk of ACL injury with non-natural surfaces. Further studies should be performed with larger sample sizes in order to further determine the risk of non-natural surfaces.

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Introduction:

The National Football League (NFL) is one of the most popular sports leagues in the United States and according to the NFL network, attracts up to 112.3 million viewers during the Super Bowl each year. American football is a high velocity, high impact sport requiring speed, size, and agility. With all that is required of these athletes, contact injuries are a prevalent aspect of the sport with a rate of .035 injuries/play[3]. These injuries have been reported by NFL analysts to cost teams up to 500 million dollars in some seasons[16]. The NFL has recently made several rule changes to help prevent injuries and improve the health of players. However, some might argue they have not considered variables that are associated with non-contact injuries, like anterior cruciate ligament injuries, which occur at an average of 62 per year and predominantly due to non-contact activities[1, 11, 12].

Anterior cruciate ligament (ACL) injuries are common and potentially devastating injuries for high level athletes. Previous studies have shown increased non-contact injuries on non-grass fields and extensive impacts on athletes including but not limited to their ability to return to play and financial cost[5]. For example, only 37% of NFL players suffering an ACL injury are able to return to play with an average time out period of 10.75 months for the ones who are able to resume their careers[14, 15]. Along with the threat of early retirement, the financial burdens of these injuries have been estimated to cost players over $2,070,521 over the four seasons following the injury with a higher effect on players who make less than $2 million per year[14]. Understanding ways to decrease the incidence of ACL tears is crucial to the physical and financial well-being of professional football players and basis for clinical research according to Mack, et al[9].

Studies have found anterior cruciate ligament injuries to be 4 times more likely to occur on artificial turf compared to natural grass[4, 10]. In September of 2020 the Associated Press published an article in which the NFL Players Association (NFLPA) called for a change to have all NFL games be played on natural grass[17]. However, at the time of this article there are still 16 NFL stadiums with either Turf or hybrid playing surfaces, not including the Canton Ohio home of the Hall of Fame game or the two stadiums in London where NFL games are beginning to be played more frequently[18]. The domestic

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prevalence in addition to the increasing popularity of internationally hosted NFL games, necessitates a more critical census of each stadium's playing surface, ACL injury rates depending on field-type, and considerations of changes to better protect the players. The mechanism of injury has been believed to be multifactorial but fatigue and what Loughran, et al. proposed as most likely due to the static position of the foot with increased traction from the field surface during athletic movements[1, 8]. The purpose of this study was to calculate the prevalence of ACL injuries in the NFL on natural grass versus non-natural grass playing surfaces and to determine if there is a significant increase on non-natural grass surfaces. We hypothesize that there will be a significant increase in ACL injuries on non-natural grass surfaces that may warrant the need to change these surfaces for future seasons.

Methods:

Sampling included the 2017/18 through the 2021/22 NFL season, for which publicly available data was reviewed to identify ACL injuries sustained by NFL athletes.This time period was chosen as a five year period which included the use of the new generation artificial playing surfaces. Publicly available data primarily included news/press releases by NFL teams and NFL game data and published injury reports (Table 1). Playing surface types were obtained via publicly available data and team websites. Included injuries were those that involved rupture or tear of the ACL and occurred in an NFL athlete in a setting in which the playing surface could be identified (Table 2). Those ACL injuries that occurred in a setting in which the playing surface could not be identified (ie practice injury or "organized team activity") were excluded. This led to the inclusion only of those ACL injuries which occurred in NFL preseason, regular season, or playoff games.

Each stadium that hosted any NFL game during our sample seasons in which an ACL injury occurred was reviewed to determine the playing surface. Playing surfaces consisted of grass, synthetic turf, and the combined synthetic/natural grass surface GrassMaster. Grass surfaces consisted of either Kentucky Bluegrass, Bermuda Grass, or a blended grass species. Turf surfaces were considered to be those with 100% synthetic playing surface without any natural grass, produced by a variety of different synthetic turf manufacturers. GrassMaster, by Tarkett Sports, is a hybrid synthetic-natural grass surface in which synthetic fibers are placed into the sub-base of the playing surface in a 2x2 cm grid fashion approximately

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18 cm deep with 1.5-2 cm exposed above ground level, then overseeded with natural grass. Only two NFL teams utilized GrassMaster during our sampled seasons, the Philadelphia Eagles (Lincoln Financial Field) and the Green Bay Packers (Lambeau Field).

Statistical Analysis:

Injury rates, defined as ACL injuries per game, were calculated for each playing surface (grass, turf, and GrassMaster) in accordance with Howard, et al[6]. Turf and GrassMaster were also analyzed as a combined group, which was deemed “non-natural” playing surfaces. Injury rate ratios of turf vs grass and non-natural surfaces vs grass were calculated. Odds ratio of turf vs grass and non-natural surfaces vs grass were calculated with 95% confidence intervals by one of the senior authors (EH) (Table 3).

Results:

During our 5 sampled seasons 2017/18 - 2021/22, gathered from publicly available data, there were 173 ACL injuries sustained by NFL athletes that occurred on an identifiable playing surface during either preseason, regular season, and playoff games. By field type, 73 on grass surfaces, 85 on turf surfaces, and 15 on Grass Master. There were a total of 1,603 games played during the sampled time period, 752 on grass, 739 on turf, 112 on Grass Master. Overall ACL injury rate was found to be 0.108 ACL injuries per game. Grass injury rate was 0.097 vs turf 0.115 vs Grass Master 0.134, with non-natural grass ACL injury rates of 0.118. Injury rate ratio to compare turf vs grass was 1.185, whereas injury rate ratio for non-natural vs grass was 1.211, representing an 18.5% increased rate of ACL injury on turf compared to grass and a 21.1% increased rate of ACL injury on non-natural compared to grass playing surfaces. Odds ratio for ACL injury comparing turf vs grass was found to be 1.209 (95%CI 0.869-1.683, p=0.261) and non-natural vs grass OR of 1.239 (95% CI 0.900-1.704, p=0.189).

Discussion:

Our study demonstrated a trend towards an increased rate of ACL injuries on non natural playing surfaces compared to natural grass. ACL injuries in the NFL continue to be a hot topic of conversation. The role of turf playing surfaces as a risk factor comes up frequently in the media and from the NFL

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Player Association. When Odell Beckham Jr. went down with an injury in Super Bowl LVI and it was revealed he had suffered an ACL rupture, the turf playing surface at SoFi stadium was the first to come under fire. Past studies on this topic have shown mixed results. Powell, et al. showed a statistically significant increase in ACL injuries on Astroturf compared to natural grass from 1980 to 1989[13]. This study however did not include more recent designs and technology in field turf or the GrassMaster playing surfaces included in our study. Drakos et al, published a paper in 2012 reviewing the relationship between playing surfaces, shoe-surface interface, and lower extremity injuries in athletes. Although most of the data observed was in high school athletes, where there may be a protective component of artificial playing surfaces. They mention a correlation of surface related injuries in elite level athletes and hypothesize the relationship of elite athletes carrying more mass and generating more power resulting in greater torque as a reason for increased non-contact injuries [19]. Balazs et al published a systematic review of 10 studies which included 963 injuries. Of these, six studies included American Football cohorts and four were soccer cohorts. Four of the studies which included American Football reached statistical significance. One of these studies included a small cohort of high school athletes and showed a statistically significant decrease in ACL injuries on turf compared to grass. Three of the studies were professional American Football. One of which showed a 92% increase in injuries on 1st generation Astroturf compared to natural grass[2]. The most relevant to our study published by Hershman et al included data from 2000-2009 which included both old generation and new generation infill turf and noted an incidence density ratio of 1.67 representing a 67% increase in rate of ACL tears on non-natural playing surfaces, which ultimately was statistically significant[10]. Our study is the first to our knowledge to include an updated investigation into the prevalence of ACL injuries in the NFL on natural grass versus non-natural playing surfaces to include the new GrassMaster playing surface which exists at both Lincoln Financial and Lambeau fields in Philadelphia and Green Bay respectively. Our study possesses some weaknesses. The most obvious weakness being that our data was collected using only publicly available sources. We reached out to the NFL to try and obtain data from the NFL's internal injury data records; however we were denied access. Inclan, et al referenced publicly obtained data (POD) studies and noted that the capture rate of ACL injuries in POD studies were 68% of true ACL injuries reported to the NFL medical staff [7]. As such it can be assumed that we have missed a number of ACL injuries that occurred,

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however it is unlikely that a greater number of either non-natural or natural grass ACL injuries were excluded due to a lack of data, and would be less likely to skew the proportions that we found in our data set. Our data is also limited to those that only occurred in preseason, regular season, or playoff game settings. Practice field surfaces were not always publicly available and most teams have both natural and non-natural practice playing surfaces. Because of this we were unable to determine the playing surface in which practice injuries occurred and these injuries were excluded. Our study demonstrated a trend towards an increased rate of ACL injuries on non natural playing surfaces compared to natural grass that was not found to be clinically significant. More studies should be performed to further evaluate the risk of ACL injuries on different playing surfaces in order to protect athletes at both the professional and amateur levels. This would be improved with collaboration with the NFL to access the NFL medical records.

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Table 1

ACL Injuries by Field Type	Total	%
Overall	173	
Grass	73	42.1
Non-Natural Field	100	57.8
Turf	85	49.1
Grassmaster	15	8.7

Table 2

Games Without ACL Injury 2017-2021	Total	%
Overall	1430	
Grass	679	47.4
Non-Natural Field	751	52.5
Turf	654	45.7
Grassmaster	97	6.8

Field Type (Table 3)	ACL Odds Ratio	95% CI Interval	p-value
Grass	0.108		

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Non-Natural Fields	0.133		
Turf	0.130		
Grass Master	0.155		
Turf vs. Grass	1.208	0.869 - 1.683	0.261
Non-Natural Vs. Grass	1.238	0.900 - 1.704	0.189

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